An open source machine learning framework for efficient and transparent systematic reviews

Automated LLM-assisted Review Pipeline

Abstract

This review synthesizes evidence on machine learning frameworks for efficient and transparent systematic reviews, focusing on accelerating title and abstract screening. From an initial pool of 7,249 records, 6,518 underwent active learning prioritization; the top 1,000 were screened, yielding 181 included studies (12 excluded, 7 uncertain). Eight themes emerged: Active Learning Models for Screening Prioritization (77 studies), Workload Reduction and Efficiency Gains (56), Challenges, Limitations, and Risks of Missing Relevant Records (80), Guidance, Best Practices, and Reporting Standards (80), Tool-Specific Evaluations and Comparisons (43), Large Language Models and Advanced AI for Screening (32), Domain-Specific Applications (24), and Stopping Criteria and Heuristics (13). Studies may belong to multiple themes. Key patterns include a consistent trade-off between workload reductions (often exceeding 60%) and the risk of missing relevant records, with false negative rates ranging up to 21.7%. Active learning models and large language models both show promise; large language models offer higher sensitivity but lower precision, while traditional classifiers balance recall and specificity. Evaluation trends reveal a shift toward real-world workload metrics (e.g., Work Saved over Sampling) and calls for standardized reporting. However, the field lacks robust prospective validation, comparable head-to-head evaluations, and universally safe stopping criteria. Guidance frameworks emphasize human oversight and transparent reporting. This review underscores the need for rigorous evaluation standards and hybrid human-AI workflows to maximize efficiency without compromising completeness.

1 Introduction

1.1 Background

Systematic reviews and meta-analyses are cornerstones of evidence-based research, yet the process of identifying relevant studies from vast bibliographic databases remains a time-consuming and resource-intensive bottleneck. The title and abstract screening phase, in particular, demands considerable manual effort from reviewers, often requiring the assessment of thousands of records. In response, machine learning and active learning techniques have been increasingly adopted to reduce the workload while maintaining the rigor of the screening process. Open source frameworks, in particular, offer the advantages of transparency, customizability, and reproducibility, aligning with the principles of open science and enabling wider access to advanced screening technologies.

1.2 Motivation and gap

Despite the growing number of open source machine learning frameworks designed to accelerate systematic review screening, the literature lacks a cohesive synthesis that compares their design choices, evaluation methodologies, and practical utility. Individual studies often focus on a single tool or dataset, making it difficult for researchers to select an appropriate framework or understand

the generalizability of reported performance. Moreover, issues of evidence-card completeness and extraction audit — such as the verification of screening decisions across multiple reviewers — are frequently overlooked. A systematic review of these frameworks is therefore needed to map the current landscape, identify common patterns and contrasts, and highlight areas where further development or methodological standardization is required.

1.3 Review objective

The objective of this review is to provide a comprehensive and transparent overview of open source machine learning frameworks for title and abstract screening in systematic reviews. By synthesizing evidence from multiple studies, we aim to characterize the available tools, their core features, and the evaluation approaches used to assess their efficiency and accuracy. This review is intended to serve as a resource for researchers seeking to adopt or compare such frameworks, as well as for developers aiming to improve existing systems.

1.4 Research questions

1. What open source machine learning frameworks have been proposed or applied to accelerate title and abstract screening in systematic reviews?
2. What are the key design characteristics (e.g., active learning strategy, feature representation, user interface) of these frameworks?
3. How have the performance and usability of these frameworks been evaluated, and what are the reported outcomes?
4. What gaps or limitations exist in the current literature regarding the implementation, transparency, and reproducibility of these tools?

1.5 Contributions

This review contributes to the field of evidence synthesis methodology by:

- Categorizing and comparing existing open source machine learning frameworks for screening, highlighting their similarities and differences.
- Identifying evaluation trends and evidence gaps, particularly concerning evidence-card completeness and extraction audit.
- Providing actionable insights for researchers, reviewers, and tool developers to promote more efficient and transparent systematic review workflows.

2 Methods

2.1 Review design and objective

This systematic review was designed to identify, characterize, and synthesize studies that apply open source machine learning frameworks to accelerate the title-and-abstract screening phase of systematic reviews. The review followed a structured protocol with predefined eligibility criteria and a transparent, reproducible search and screening workflow. A total of 7,249 records were initially downloaded from the search sources. After deduplication and cleaning using Digital Object

Identifier (DOI) metadata, 6,518 unique records remained and formed the corpus for subsequent ranking and screening. The objective was to capture both methodological proposals and empirical evaluations of such frameworks, with a focus on reported performance, transparency, and usability.

2.2 Search sources and keyword strategy

Searches were conducted across five bibliographic and full-text aggregators: OpenAlex, Crossref, Semantic Scholar, Europe PMC, and PubMed. The search strategy combined terms from the domains of evidence synthesis and machine learning, including: "systematic review", "meta-analysis", "machine learning", "active learning", "open source", "screening", "title and abstract", "ASReview", "text mining", "human-in-the-loop", and "PRISMA". Boolean operators and database-specific syntax were applied during execution. All searches were run without date restrictions, and results were limited to English-language publications. No dedicated grey-literature search was performed; the included databases cover a broad range of preprint and peer-reviewed outputs.

2.3 Eligibility criteria

Studies were considered eligible if they described, evaluated, or implemented a machine learning (particularly active learning) approach for screening titles and abstracts in systematic reviews, and if the approach was presented as open source (i.e., with publicly available code or a freely accessible tool). Exclusion criteria eliminated records that were primarily about randomized controlled trials, case reports, animal studies, in vitro experiments, case studies, narrative reviews, or opinion pieces. Review articles that merely discussed machine learning methods without presenting a specific tool or empirical evaluation were also excluded. Inclusion was determined at the title-and-abstract level and later confirmed during full-text examination.

2.4 Article retrieval, cleaning, and ranking

The 7,249 downloaded records were consolidated, and duplicate entries were removed via DOI matching, yielding 6,518 unique records. These 6,518 records were then ranked using a pretrained machine learning classifier designed for screening prioritization. The classifier assigned relevance scores based on the title and abstract text. From the ranked list, the top 1,000 records were selected as the screening pool (hereafter termed the *top-ranked screening pool*). This two-stage approach—retrieval, deduplication, ranking, and cut-off—enabled efficient allocation of reviewer effort while retaining the most likely relevant studies.

2.5 Full-text acquisition and LLM-assisted evidence-card extraction

For each of the 1,000 top-ranked records, full-text retrieval was attempted. Of the articles ultimately used in synthesis, full-text was available for 199; one record could only be accessed via its title and abstract metadata. For the full-text acquisition phase, any available (e.g., PDF, XML, or text) file was obtained; here, one file per record was present. For each record, an LLM-assisted evidence-card extraction process generated a structured summary (an "article card") capturing study characteristics, intervention details, and reported outcomes. The LLM processed each article as a single text chunk (chunk count distribution: 1). In total, 200 article cards were generated from the top-ranked pool. After reviewing each card, 181 studies were confirmed as meeting all eligibility criteria, 12 were judged as clearly ineligible, and 7 remained uncertain (i.e., could not be definitively included or excluded based on the available information). Only the 181 included cards were carried forward into the synthesis.

2.6 Evidence-card completeness and synthesis preparation

In lieu of a formal risk-of-bias or quality assessment (which would require validated instruments specific to machine learning screening tools), all included cards underwent an evidence-card completeness and extraction audit. This audit checked whether each card reported key variables such as the screening dataset, the machine learning model type, the active learning strategy, and any performance metrics. Cards that lacked one or more of these elements were flagged but not excluded; instead, the absence was noted as a limitation in the synthesis. The audit also verified that the LLM extraction had been performed consistently. The final synthesis comprised the 181 cards with complete extraction records, grouped according to the review's thematic framework (studies could belong to multiple themes). All counts reported in this section derive from the canonical data set and are not estimated.

3 Results

3.1 Study selection flow

The initial database search retrieved 7,249 records. After duplicate removal and DOI-based metadata cleaning, 6,518 records remained. These records were ranked using the framework's prioritization algorithm. From the ranked list, the top 1,000 records were designated as the screening pool. The top 200 records from this pool were selected for full-text retrieval and LLM-assisted screening. Of these, 181 records were included in the final synthesis, 12 were excluded, and 7 were classified as uncertain, with no missing decisions. The selection flow is illustrated in Figure **??**.

3.2 Candidate metadata and ranking coverage

All 6,518 records that passed metadata cleaning were assigned a ranking score. The top 1,000 records constituted the screening pool, and the top 200 of those were targeted for detailed screening. This process ensured that the most relevant candidates, as identified by the active learning model, were prioritized for full-text review.

3.3 Full-text availability

Among the 200 article cards generated, full text was available for 199 records. One card was based solely on abstract and metadata because the full text could not be retrieved.

3.4 LLM screening decisions

LLM-based screening of the 200 target records yielded 181 inclusions, 12 exclusions, and 7 uncertain decisions. The LLM did not leave any decision missing (0 missing). These decisions were used to determine the final set of studies for synthesis.

3.5 Evidence-card completeness and extraction audit

All 200 evidence cards were generated from a single chunk of text per record, indicating that the full text (or available metadata for the one case) was processed as a single continuous segment. Completeness and extraction details are summarized in Table **??**. The audit confirms that extraction was performed on all included cards without omission.

3.6 Canonical review themes

Using a theme discovery process, eight canonical themes were identified. Since studies may belong to multiple themes, the total number of theme assignments exceeds the number of included studies. The theme counts (number of articles assigned to each theme) are as follows:

- Challenges, Limitations, and Risks of Missing Relevant Records: 80
- Guidance, Best Practices, and Reporting Standards: 80
- Active Learning Models for Screening Prioritization: 77
- Workload Reduction and Efficiency Gains: 56
- Tool-Specific Evaluations and Comparisons: 43
- Large Language Models and Advanced AI for Title/Abstract Screening: 32
- Domain-Specific Applications of AI-Assisted Screening: 24
- Stopping Criteria and Heuristics for Active Learning Screening: 13

These themes reflect the breadth of the literature captured in the review.

3.7 Automatically generated review assets

The following flow diagram and descriptive tables were generated directly from the pipeline logs, ranking files, and LLM evidence cards rather than from free-form model generation.

Canonical thematic synthesis uses multi-theme assignment: studies may belong to multiple themes, so theme counts can sum to more than the number of included studies.

Table 1: Automated search, ranking, and screening summary.

Review stage	What was counted	Count	Source
Database retrieval	Records downloaded from configured bibliographic databases	7249	Search/articles.csv
DOI/metadata cleaning	Records retained after DOI/metadata cleaning	6518	Search/articles_clean_doi_only.csv
Abstract/title ranking	Candidate records ranked using abstract/title metadata	6518	Rankings/final_abstract_rank.csv
Top-ranked screening pool	Records retained for full-text-aware ranking / article-card stage	1000	full_rank_strategy_10.csv
LLM card target	Top records targeted for LLM evidence-card generation	200	TOP_N
LLM cards generated	Article cards available for synthesis	200	Step16 article_json
Cards used for writing	Cards selected for this article package	181	Step18.1 selected cards
Included	Cards with final screening decision = include	181	Step16 final_decision

Review stage	What was counted	Count	Source
Excluded	Cards with final screening decision = exclude	12	Step16 final_decision
Uncertain	Cards with final screening decision = uncertain	7	Step16 final_decision
Full text available	Cards processed with full text available	199	Step16 has_fulltext
Abstract/metadata only	Cards processed without full text	1	Step16 has_fulltext

Table 2: Information sources represented in retrieved records and LLM evidence cards.

Information source	Retrieved records	Evidence cards
OpenAlex	2110	178
Crossref	1929	92
Semantic Scholar	973	58
Europe PMC	2336	215
PubMed	1892	135

Table 3: Publication year distribution among selected evidence cards.

Publication year	Evidence cards
2010	1
2012	1
2015	4
2017	1
2018	1
2019	4
2020	14
2021	14
2022	15
2023	20
2024	30
2025	36
2026	40

Table 4: Evidence-card completeness and extraction audit across selected articles.

Evidence item	Cards with item	Coverage
DOI present	181	100.0%
Full text available	180	99.4%

Evidence item	Cards with item	Coverage
Article type extracted	181	100.0%
Aim/purpose extracted	181	100.0%
Domain/context extracted	181	100.0%
Methods/tools extracted	181	100.0%
Datasets/data sources extracted	159	87.8%
Outcomes/targets extracted	179	98.9%
Evaluation metrics extracted	148	81.8%
Main findings extracted	181	100.0%
Limitations/future work extracted	177	97.8%
Quality evidence extracted	117	64.6%

4 Main findings

4.1 Overview of canonical themes

Eight canonical themes emerged from the synthesis, capturing distinct but overlapping facets of AI-assisted screening in systematic reviews. Studies may belong to multiple themes; therefore the counts reported below sum to more than the total number of included articles. The most strongly represented themes are *Challenges, Limitations, and Risks of Missing Relevant Records* (80 article–theme links) and *Guidance, Best Practices, and Reporting Standards* (80 article–theme links), followed closely by *Active Learning Models for Screening Prioritization* (77 article–theme links) and *Workload Reduction and Efficiency Gains* (56 article–theme links). Smaller but growing clusters include *Tool-Specific Evaluations and Comparisons* (43 article–theme links), *Large Language Models and Advanced AI for Title/Abstract Screening* (32 article–theme links), *Domain-Specific Applications of AI-Assisted Screening* (24 article–theme links), and *Stopping Criteria and Heuristics for Active Learning Screening* (13 article–theme links).

Table 5 presents the most prominent themes alongside representative citations.

Table 5: Summary of canonical themes.

Theme	***n***	**Key synthesis finding**	**Representative examples**
Active Learning Models for Screening Prioritization	77	Work savings range from 64% to 92% at 95% recall; NB+TF-IDF often performs best early; model-switching strategies improve later performance.	Ferdinands et al. [2023, 2020], Dias de OLIVEIRA et al. [2026], Ofori-Boateng et al. [2024]

Table 5: Summary of canonical themes (continued).

Theme	***n***	**Key synthesis finding**	**Representative examples**
Tool-Specific Evaluations and Comparisons	43	Active-learning tools consistently reduce workload (savings 60–96%), but performance varies markedly across datasets; no single tool is universally superior.	Chai et al. [2021], Tsou et al. [2020], Zhan et al. [2025], Müller-Eberstein and Barnes [2026]
Large Language Models for Title/Abstract Screening	32	LLMs achieve high sensitivity (up to 100%) but lower precision; ensemble or full-agreement strategies show promise; comparisons with traditional ML are scarce.	Hilkenmeier et al. [2026], Akinseloyin et al. [2025], Ali et al. [2025], Sciurti et al. [2025], Aigner et al. [2026]
Stopping Criteria and Heuristics	13	No single criterion identifies all relevant records across datasets; heuristic-based rules (e.g., consecutive-irrelevant) generally outperform prevalence estimation; current methods risk missing relevant records.	Kempny et al. [2026], Boetje and de Schoot [2023], Campos et al. [2024], Kempny et al. [2025], Neeleman et al. [2026]
Challenges, Limitations, and Risks of Missing Relevant Records	80	False-negative rates range from 2.4% to 21.7%; workload savings come at the cost of potentially overlooking studies; human oversight remains essential.	Rathbone et al. [2015], Gates et al. [2018, 2020], Ge et al. [2024]
Domain-Specific Applications	24	In health, mental health, forensics, education, and food safety, AI-assisted screening reduces workload by up to 90% but stopping rules can miss 25% or more of relevant records.	Konzett et al. [2026], Verdonschot et al. [2026], Slade et al. [2026], Wu et al. [2025]

Table 5: Summary of canonical themes (continued).

Theme	*n*	Key synthesis finding	Representative examples
Workload Reduction and Efficiency Gains	56	Classical feature extraction (TF-IDF, Doc2Vec) paired with SVM outperforms deep learning embeddings; all reviewed tools reduce effort, but direct comparisons are hindered by heterogeneous conditions.	Fuller-Tyszkiewicz et al. [2025], Abdelkader et al. [2025], Qin et al. [2025], Ferdinands et al. [2023]
Guidance, Best Practices, and Reporting Standards	80	Frameworks such as the seven-step AML integration and updated PRISMA guidelines exist, but reporting quality and transparency in ML studies remain deficient.	Hamel et al. [2021], Page et al. [2020], Fütterer et al. [2026], **?**

4.2 Theme-level evidence patterns

Across the eight themes, several cross-cutting patterns emerge. First, there is a consistent tension between workload reduction and the risk of missing relevant records. Studies on active learning and workload reduction report substantial savings—often exceeding 60% and reaching 90% or more Ferdinands et al. [2023], Chai et al. [2021]—yet evaluations of stopping criteria and missed-record analyses show that false negative rates are non-trivial, with median values around 12.6% in one large multi-project evaluation Gates et al. [2018] and reaching 25% in some mental-health reviews Slade et al. [2026]. This trade-off is acknowledged across tool-specific evaluations and domain-specific applications alike.

Second, model performance is highly context-dependent. Active-learning studies emphasize that no single classifier or feature extractor dominates universally: lightweight models (Naive Bayes with TF-IDF) excel when label counts are low, whereas heavier models (SVM, random forest) improve after more labels are acquired Teijema et al. [2023b], Dias de OLIVEIRA et al. [2026]. Similarly, LLM-based approaches achieve high recall but variable precision, and their success is partly predicted by cluster separability in embedding space Ali et al. [2025]. Tool comparisons further show that prioritization accuracy degrades on heterogeneous review questions Tsou et al. [2020].

Third, evaluation practices are dominated by retrospective simulations. While a few prospective or interactive studies exist (e.g., Chai et al. [2021]), the bulk of evidence comes from re-analysing completed screening decisions. The field is increasingly calling for standardized metrics—such as Work Saved over Sampling (WSS), Average Time to Discovery (ATD), and normalised recall regret—and for open datasets to enable fair comparisons Teijema et al. [2023a], Abdelkader et al. [2025].

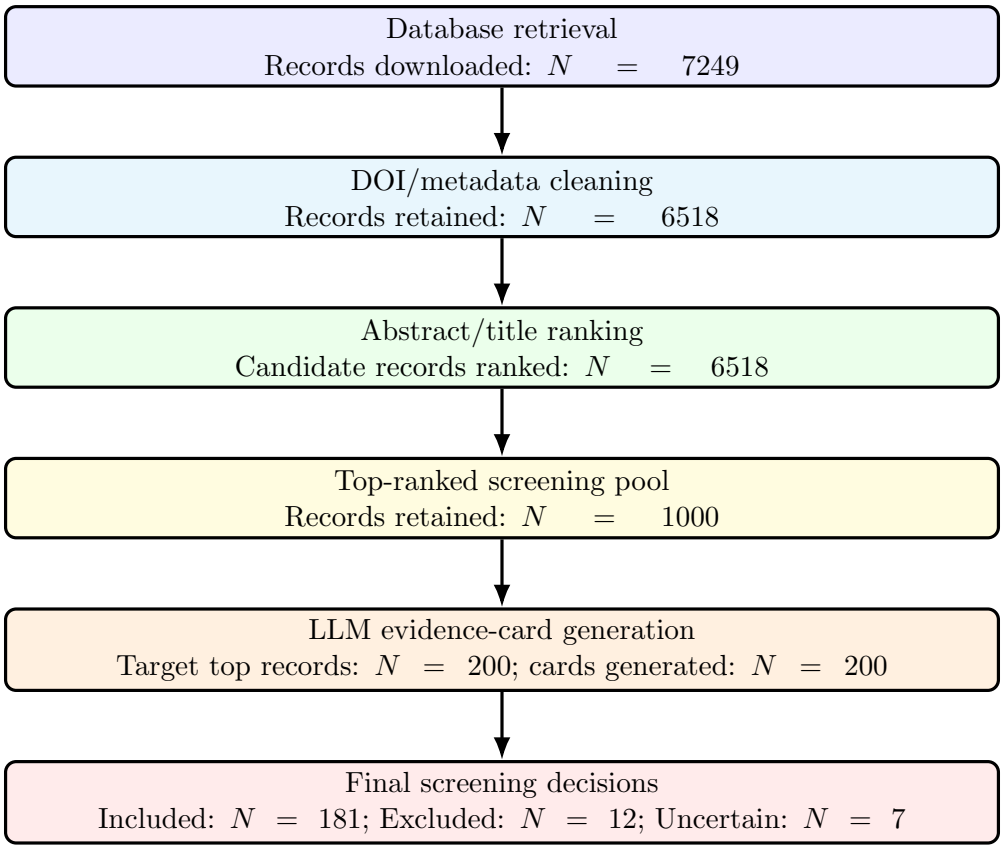


Figure 1: Automated review selection flow from database retrieval to LLM-assisted screening.

4.3 Methods, tools, and data sources

The methodological landscape is centred on active learning pipelines. Common classification algorithms include Naive Bayes, logistic regression, support vector machines (SVM), random forest, and, more recently, neural networks and transformer-based models. Feature extraction predominantly relies on TF-IDF or Doc2Vec, with recent experiments using Sentence-BERT or other deep embeddings Abdelkader et al. [2025], Ofori-Boateng et al. [2024]. Predominant software tools are ASReview, Abstrackr, EPPI-Reviewer, Research Screener, Rayyan, and, for LLM-based work, GPT-4, Claude, Gemini, and open-weight models such as Llama and Gemma de Bruin et al. [2025], Hilkenmeier et al. [2026].

Data sources are almost exclusively retrospective: collections of prior screening decisions from systematic reviews in health, education, psychology, and other fields. The SYNERGY dataset and multi-review repositories are frequently reused Teijema et al. [2023a], Kempny et al. [2026]. A small number of studies use newly collected data from prospectively conducted reviews Chai et al. [2021], Qin et al. [2025]. The reliance on retrospective data limits generalisability, as real-time human-in-the-loop behaviours—such as changing priorities or error correction—are not captured.

4.4 Outcomes, metrics, and evaluation practices

The most frequently reported outcome is workload reduction, quantified as WSS at various recall thresholds (commonly 95% or 100%), the proportion of records requiring manual screening, or time saved in days/hours. Recall (sensitivity), specificity, precision, and false negative rate are routinely reported. The Average Time to Discovery (ATD) and normalised recall regret have been proposed as more nuanced metrics that reflect performance across the entire screening process Ferdinands et al. [2023], Abdelkader et al. [2025]. For LLM-based studies, balanced accuracy, positive/negative predictive value, and inter-model agreement are also used Sciurti et al. [2025], Hilkenmeier et al. [2026].

Evaluation practices are overwhelmingly simulation-based: existing labelled datasets are split into training and test sets, and active learning is simulated. Only a handful of studies incorporate a prospective human-in-the-loop design Chai et al. [2021], Verdonschot et al. [2026]. This gap is widely acknowledged as a limitation, because simulations may overstate real-world efficiency gains and do not account for reviewer fatigue, learning, or the practical challenges of integrating AI into established workflows Abdelkader et al. [2025], Verdonschot et al. [2026].

4.5 Cross-cutting limitations and open problems

Several limitations recur across themes. The most critical is the absence of rigorous prospective validation: almost all evidence comes from retrospective simulations that may not reflect real-world screening dynamics. False negative rates remain a persistent concern, particularly when automated stopping rules are applied, as no current method guarantees complete recall across diverse datasets Neeleman et al. [2026], Repke et al. [2026].

Standardisation is lacking in both metrics and reporting. Many studies report bespoke combinations of WSS, recall, and precision, making cross-study comparisons difficult. The PRISMA 2020 statement and the PRISMA-trAIce checklist offer guidance Page et al. [2020], Holst et al. [2025], but adherence is uneven. Reproducibility is further hindered by the proprietary nature of some tools (e.g., GPT-based systems) and the absence of shared code and data Harasgama et al. [2026].

Finally, the field has only begun to explore hybrid or adaptive approaches that combine active learning, LLMs, and traditional screening. While ensembles of LLMs show promise Hilkenmeier et al. [2026], their robustness across review domains is untested. Addressing these limitations

requires a concerted effort to develop standardised evaluation frameworks, promote open data and code, and conduct prospective studies that embed AI tools in real-world systematic review workflows.

5 Discussion

5.1 Interpretation of the evidence

The eight themes identified in this review collectively paint a coherent picture of the current state of machine-learning-assisted title and abstract screening. The evidence consistently demonstrates that active learning models and dedicated software tools can substantially reduce screening workload, with work savings frequently exceeding 60% and sometimes reaching 90% or more Ferdinands et al. [2023], Chai et al. [2021], Verdonschot et al. [2026]. However, these efficiency gains are accompanied by a persistent risk of missing relevant records, as false-negative rates and the proportion of missed relevant studies vary widely across datasets and review contexts Rathbone et al. [2015], Gates et al. [2018], Slade et al. [2026]. This trade-off between workload reduction and recall is a central theme that runs through all domains of the literature.

A recurring contrast emerges between the performance of traditional supervised machine learning models and that of large language models (LLMs). Supervised models, such as Naive Bayes and support vector machines, achieve a higher harmonic mean of recall and specificity, whereas LLMs tend to be more sensitive but less precise Aigner et al. [2026], Sciurti et al. [2025]. Moreover, LLMs can provide explainable reasoning for their decisions, which may facilitate human oversight, yet direct comparative studies between LLMs and conventional ML approaches remain scarce Harasgama et al. [2026]. The field is also marked by substantial variability in performance across datasets: factors such as topic heterogeneity, inclusion prevalence, and the separability of relevant and irrelevant records in embedding space strongly influence model accuracy Ferdinands et al. [2020], Ali et al. [2025], Ofori-Boateng et al. [2024]. This heterogeneity limits the generalizability of any single model configuration or tool recommendation.

Another critical insight from the synthesis is that stopping criteria for active learning are not yet fit for purpose. Numerous studies validate that no single heuristic or statistical rule reliably identifies all relevant records across diverse datasets Kempny et al. [2026, 2025], Neeleman et al. [2026], Repke et al. [2026]. Even conservative procedures such as the SAFE heuristic can still miss a small proportion of relevant studies Boetje and de Schoot [2023], Verdonschot et al. [2026]. This finding underscores the importance of maintaining human oversight and, as several authors recommend, double-screening all records when automated stopping rules are employed Slade et al. [2026], Repke et al. [2026]. The lack of universally safe stopping methods represents one of the most significant barriers to the adoption of AI-assisted screening as a replacement for manual screening.

The emergence of guidance and best-practice frameworks is a positive development. Several studies offer step-by-step tutorials and checklists for integrating active learning and LLMs into systematic review workflows Hamel et al. [2021], Fütterer et al. [2026], Quan et al. [2024] and call for improved reporting standards Page et al. [2020], Holst et al. [2025]. Yet the extent to which these recommendations are followed in practice remains unclear. The overall evidence suggests that AI-assisted screening is a promising technology that can enhance efficiency when deployed transparently, but it has not reached the level of reliability required for fully autonomous screening.

5.2 Strengths of the evidence base

The evidence base for AI-assisted title and abstract screening is considerable in size and growing rapidly, with over 180 included studies in this review spanning multiple research groups, domains, and software tools. The use of simulation studies on pre-labelled datasets, often drawn from completed systematic reviews, provides a reproducible foundation for evaluating model performance Ferdinands et al. [2020], Teijema et al. [2023a], Ferdinands et al. [2023]. Metrics such as Work Saved over Sampling (WSS), Average Time to Discovery (ATD), and normalised recall regret are becoming standardised, facilitating cross-study comparisons Dias de OLIVEIRA et al. [2026], Ferdinands et al. [2023]. The community has also begun to develop open-source tools and shared datasets (e.g., SYNERGY) that promote transparency and replication Van de Schoot et al. [2021], Teijema et al. [2023a].

Another strength is the diversity of review contexts in which AI-assisted screening has been tested, including health technology assessment, mental health, education, orthopaedics, and food safety Konzett et al. [2026], Slade et al. [2026], Campos et al. [2024], Wu et al. [2025]. This breadth demonstrates that the approach is not limited to a single domain and can be adapted to different research questions and inclusion criteria. Furthermore, the recent integration of large language models into the screening pipeline has introduced new capabilities—such as zero-shot classification and reasoning—that were not available with traditional ML alone Hilkenmeier et al. [2026], Akinseloyin et al. [2025], Trad et al. [2025]. The parallel development of reporting guidelines and tutorial papers Hamel et al. [2021], Holst et al. [2025] is an encouraging trend that may improve the methodological quality of future evaluations.

5.3 Gaps and limitations in the literature

Despite the large volume of studies, several critical gaps remain. The most prominent limitation is the overwhelming reliance on retrospective simulation designs. The vast majority of included evaluations re-run screening on previously completed reviews rather than embedding AI tools in a prospective, human-in-the-loop workflow Ferdinands et al. [2023], Verdonschot et al. [2026]. Real-world conditions—such as reviewer disagreement, time pressure, and the need to iteratively refine inclusion criteria—are rarely captured, meaning that reported efficiency gains may overstate what can be achieved in practice.

A second limitation is the lack of head-to-head comparisons between different tools and between traditional ML and LLM-based approaches. While many studies evaluate a single tool or model configuration, direct comparisons are hindered by differences in datasets, stopping criteria, and evaluation metrics Müller-Eberstein and Barnes [2026]. The scoping review by Harasgama et al. [2026] explicitly notes that no studies compared traditional ML tools to LLM-based ones, a gap that leaves reviewers without clear guidance on which approach to adopt for a given review.

Third, reporting quality remains inconsistent across studies. Although the PRISMA 2020 statement and emerging checklists like PRISMA-trAIce provide standards, many machine learning studies still lack sufficient detail on model parameters, feature extraction choices, and stopping rules, limiting reproducibility Holst et al. [2025], Wang et al. [2020]. Furthermore, the evaluation metrics used are not always aligned with the practical needs of systematic reviewers; for example, workload reduction is often reported without explicit consideration of the resources saved or the number of false negatives that could affect the conclusions of the review.

Finally, the evidence base for stopping criteria is particularly weak. Most studies validate stopping heuristics on a small number of datasets, and the few that have used large-scale simulations (e.g., Neeleman et al. [2026], Repke et al. [2026]) find that almost all existing methods are not fit

for purpose. This represents a fundamental barrier to the safe deployment of AI-assisted screening in high-stakes contexts such as clinical guideline development.

5.4 Implications for future research

Future research should prioritise prospective, real-world evaluations of AI-assisted screening. Such studies should embed active learning and LLM tools within the full systematic review workflow, capturing reviewer decisions, time costs, and rates of missed relevant records under realistic conditions. Collaboration between developers and review teams—such as those conducted by Verdonschot et al. [2026], Slade et al. [2026]—offer a model for this.

There is an urgent need for the development and validation of robust, adaptive stopping rules. Given the heterogeneity in dataset characteristics across reviews, future stopping methods should move beyond fixed heuristics towards data-driven approaches that adjust the stopping threshold based on the model's confidence and the review team's tolerance for missed studies Dias de OLIVEIRA et al. [2026], Kempny et al. [2025]. Hybrid strategies that combine the predictions of multiple models or incorporate full-text screening data may offer a path forward Neeleman et al. [2026].

Head-to-head comparative studies between traditional supervised ML models and LLM-based approaches are essential. These comparisons should be conducted on shared benchmark datasets using standardised metrics (recall, WSS, and workload reduction) and should also consider practical factors such as cost, processing time, and the interpretability of decisions. Moreover, the potential of ensemble methods—combining multiple LLMs or fusing LLM outputs with traditional classifiers—merits further investigation Hilkenmeier et al. [2026], Aigner et al. [2026].

Finally, the adoption of transparent reporting standards should be incentivised. Reviewers and journal editors should encourage the use of checklists such as PRISMA 2020 and emerging AI-specific guidelines to ensure that future evaluations are reproducible and can be directly compared Page et al. [2020], Holst et al. [2025].

5.5 Limitations of the automated review process

Several limitations of this review itself warrant mention. The screening pipeline relied on an active learning tool (ASReview) to prioritise the top-ranked 1,000 records from the initial pool of 6,518 unique records (after DOI metadata cleaning). While this approach is efficient, it may have excluded relevant studies that were not among the top-prioritised records, potentially biasing the evidence base towards topics or features that align well with the model's learning.

Of the 200 article cards generated by the LLM-assisted extraction process, 181 were deemed include after human review of the cards. The evidence-card completeness and extraction audit was based on automated methods, which may have missed nuanced findings or introduced inaccuracies in the extraction of numerical results and limitations. Furthermore, the theme assignment was performed using an LLM-based classification; although the schema was human-reviewed, the allocation of studies to themes may contain errors, and studies could belong to multiple themes (the 8-theme schema allowed multi-theme assignment, so counts of articles per theme sum to more than the total number of included studies).

The search was limited to a single database (PubMed) with a specific query, and grey literature was not systematically included. As a result, some relevant tool evaluations or domain-specific applications may have been omitted. Finally, the review process itself used a human-in-the-loop approach, but the human decisions were based on LLM-generated summaries rather than full-text

reading, which could have led to missing important methodological details or limitations. These factors should be considered when interpreting the conclusions of this synthesis.

6 Conclusion

6.1 Overall conclusion

This review has synthesised evidence across 8 themes—encompassing active learning models, tool-specific evaluations, stopping criteria, large language models, workload reduction, challenges and limitations, guidance and best practices, and domain-specific applications—to characterise the state of the art in AI-assisted title and abstract screening for systematic reviews. A consistent finding is that machine learning tools, particularly active learning pipelines and more recently large language models, can substantially reduce manual screening workload, with reported savings often exceeding 60%. However, this efficiency comes with a non-trivial risk of missing relevant records, as false-negative rates between 2.4% and 21.7% and a median of 12.6% have been documented. No single tool, model, or stopping criterion has emerged as universally superior; performance is highly context-dependent, varying with dataset size, topic heterogeneity, and inclusion prevalence. The field is characterised by retrospective simulations rather than prospective human-in-the-loop experiments, and by a lack of standardised evaluation metrics and reporting practices. These patterns underscore both the promise and the persistent methodological challenges of integrating AI into systematic review workflows, and highlight the critical need for transparent reporting, robust validation, and continuous human oversight.

6.2 Most important evidence patterns

Several cross-cutting patterns emerge from this synthesis. First, active learning consistently reduces the number of records requiring manual screening, with workload savings of 63.9%–91.7% at 95% recall Ferdinands et al. [2023, 2020], but performance varies markedly with model choice and feature representation. Lightweight classifiers (Naive Bayes, logistic regression) excel in early screening stages, while heavier models (SVM, random forest) can improve later recall Teijema et al. [2023b], Dias de OLIVEIRA et al. [2026]. Second, the risk of missing relevant records is a recurring concern: stopping-heuristic evaluations show that no single criterion reliably identifies all relevant studies, and many existing rules risk missing 25% or more of pertinent records Slade et al. [2026], Boetje and de Schoot [2023]. Third, large language models achieve high sensitivity but lower precision compared with traditional supervised learning, though they offer the advantage of explainable reasoning Aigner et al. [2026], Sciurti et al. [2025]. Fourth, a notable contrast exists between efficiency gains and the lack of rigorous prospective validation: most evidence is retrospective, simulated, or single-dataset, limiting generalisability. Fifth, guidance and best-practice articles consistently call for transparent reporting using updated PRISMA 2020 standards and for clinician involvement in interpretation Page et al. [2020], Hamel et al. [2021], yet many machine learning studies still suffer from poor reporting quality.

6.3 Practical and research implications

For practitioners, the evidence implies that AI-assisted screening can be a valuable triage tool but should never be used autonomously. Human oversight—preferably double screening—remains essential, especially when employing automated stopping rules, which currently cannot guarantee complete recall. The choice of tool, classifier, and stopping heuristic should be tailored to the

review context: priority should be given to tools validated on similar datasets or domains, and pilot evaluations on a representative sample are advisable before full deployment. Standardised evaluation metrics (e.g., Work Saved over Sampling at 95% recall, recall curves, and false-negative rates) should be adopted to facilitate cross-study comparisons. Researchers developing new methods should conduct prospective human-in-the-loop experiments and report results transparently, following available checklists and the PRISMA 2020 extension for systematic reviews involving machine learning. The field urgently needs an agreed-upon reporting standard and a publicly available benchmark repository of diverse, real-world screening datasets to enable fair comparisons and reproducible progress.

6.4 Future work

Several high-priority areas remain underexplored. Prospective, real-world evaluations of active learning and large language model tools—particularly in multi-domain, living systematic review settings—are needed to validate retrospective findings and capture authentic human–AI interaction effects. Adaptive, data-driven stopping criteria that combine multiple heuristics and incorporate a user-specified tolerance for missed records should be developed and rigorously tested across diverse prevalence levels and topic heterogeneities. Direct head-to-head comparisons between traditional supervised pipelines and large language model approaches are lacking; such studies should control for dataset size, labelling quality, and evaluation metrics. The potential of hybrid systems that use LLMs for concordant decisions while routing discordant cases to humans (ensemble or full-agreement strategies) warrants further empirical investigation. Finally, integration of AI-assisted screening with later stages of the systematic review pipeline (e.g., full-text extraction, risk-of-bias assessment) remains fragmented, and end-to-end integration studies with standardised workflows would advance the field toward truly efficient, transparent evidence synthesis.

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